a) $\lim_{x \to 1^+} \frac{1^+}{x^2-1^+} = \infty$ infinito!

$x \to 1^+,\ x > 1 \Rightarrow x^2 - 1 > 0$

e) $\lim_{x \to 1^-} \frac{1^+}{x^2-1^-} = -\infty$

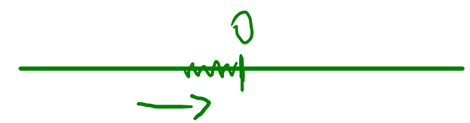
$x \to 1^-,\ x < 1 \Rightarrow x^2 - 1 < 0$

c) O lim é uma indeterminação $\frac{0}{0}$, mas ao calcular seus lim laterais foram obtidos os seguintes intervalos.

Para 1^+

$x > 0$

Para 1^-



$x < 0$

$$\lim_{x \to 2^+} \frac{x+1}{(x-2)^2} \begin{matrix} + \\ + \end{matrix} = \infty \quad x \to 2^+, \; x > 2 \Rightarrow (x-2)^2 > 0$$

$$x + 1 > 0$$

$$\lim_{x \to 2^-} \frac{x+1}{(x-2)^2} \begin{matrix} + \\ - \end{matrix} = -\infty \quad x \to 2^-, \; x < 2 \Rightarrow (x-2)^2 < 0$$

$$x + 1 > 0$$

8) $f(x) = \begin{cases} x+1, & x<0 \\ 2x+1, & 0 \leq x < 1 \\ 2+x^2 & x \geq 1 \end{cases}$

$\lim_{x \to 0} f(x) \Rightarrow \lim_{x \to 0} x+1 = 1$

$\lim_{x \to 2} f(x) \Rightarrow \lim_{x \to 2} 2 + \cancel{x^2}^{4} = 6$

$\lim_{x \to 1} f(x) \Rightarrow \lim_{x \to 1} 2 + x^2 = 3$

h) Calcular $\lim_{x \to 5^+} \frac{x-5}{|x-5|}$ e $\lim_{x \to 5^-} \frac{x-5}{|x-5|}$

$$|x| = \begin{cases} x ; x \geq 0 \\ -x ; x < 0 \end{cases}$$

$x \to 5^+$, $x > 5 \Rightarrow |x-5| > 0$

$\hookrightarrow x - 5 > 0$

$$\lim_{x \to 5^+} \frac{x-5}{|x-5|} \begin{matrix} + \\ + \end{matrix} = \infty$$

$x \to 5^-$, $x < 5 \Rightarrow |x-5| = -(x-5) = -x+5 > 0$

$\hookrightarrow x - 5 < 0$

$$\lim_{x \to 5^-} \frac{x-5}{|x-5|} \begin{matrix} - \\ + \end{matrix} = -\infty$$